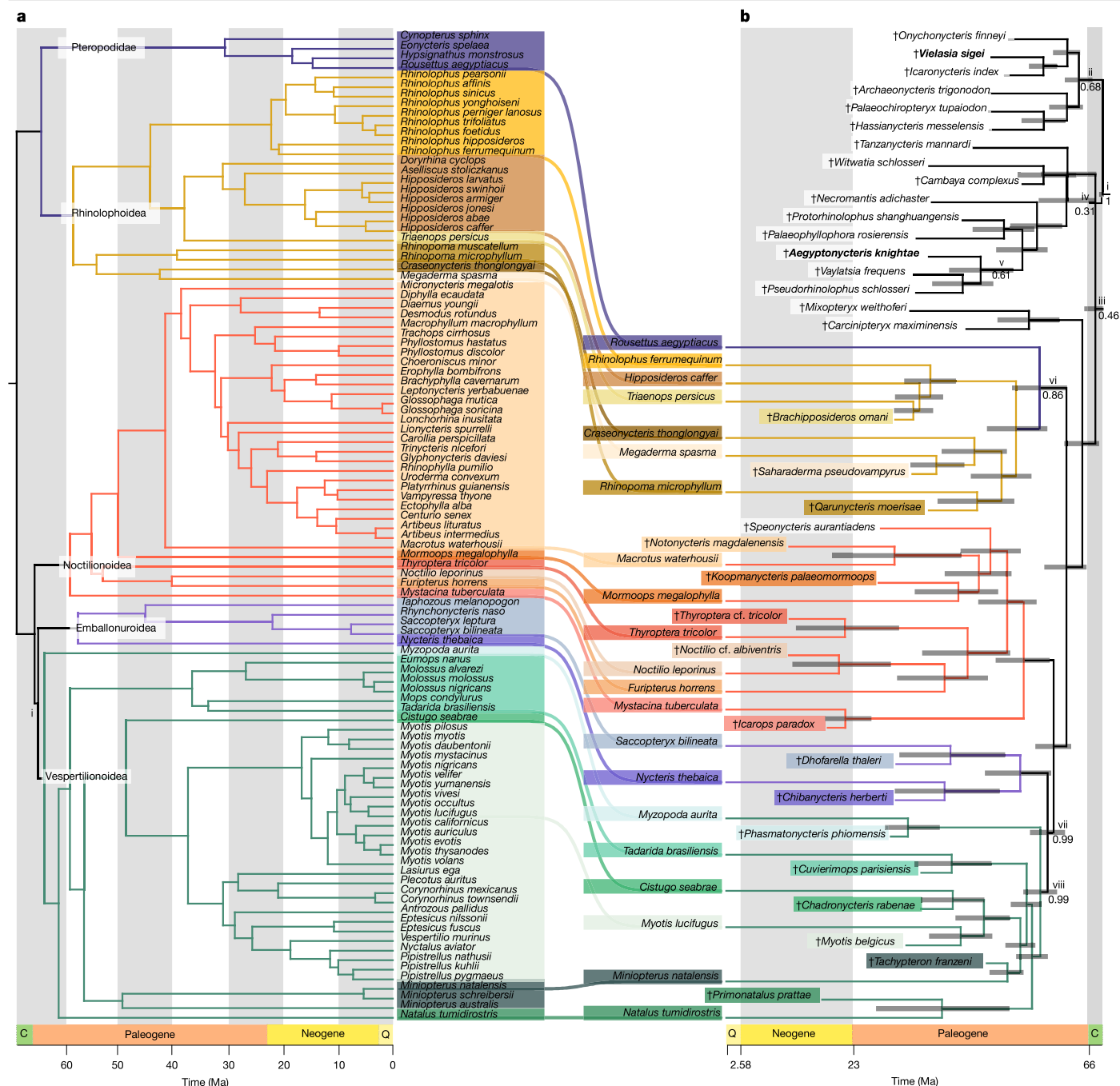




**Fig. 2 | Timetrees based on node-based and tip-dating approaches with geological timescales. a**, Mean divergence times from Bayesian analyses of coding sequences and fossil-based constraints on the T2 tree. The sister relationship between the superfamilies Emballonuroidea and Vespertilionoidea is also shown (i). C, Cretaceous; Q, Quaternary. **b**, Divergence times based on total evidence dating, including fossils as tips under the fossilized birth–death process, or tip dating, with posterior probabilities of key node support and 95% high probability density interval of posterior node age: (i) Chiropteran

within Noctilionoidea (M + N) and 11% supporting a sister relationship with Emballonuroidea (M + E) (Fig. 3b). No analyses supported an early branching position in Yangochiroptera. Using gene concordance factor (gCF) analyses, which quantifies gene tree concordance–discordance, we assessed the level of concordance per gene and found similar support for the alternate positions of Myzopodidae (gCF: 22.89% (M + V), 16.74% (M + N), 10.18% (M + E) and 50.19% (other); Supplementary Note 11). Only 22% of loci supported the position of Myzopodidae as

root placement. (ii) Oldest branching bat clade, Echioptera, including the laryngeal echolocating species **†Vielasia** highlighted in bold. (iii) Unnamed clade comprising the sister to the redefined Echioptera. (iv) Unnamed clade comprising both laryngeal echolocating species and the oldest bat inferred to feed on plants, **†Aegyptonycteris**, highlighted in bold. (v) Unnamed and smallest clade. (vi) Yinpterochiroptera. (vii) The sister relationship between the superfamilies Emballonuroidea and Vespertilionoidea, and (viii) Vespertilionoidea, including Myzopodidae.

sister to Vespertilionoidea (M + V). This placement received higher support than any alternative (Supplementary Table 11.1 and Supplementary Fig. 11.1). These results reveal extensive discordance among protein-coding genes, most likely driven by the varying selection pressure on the functional proteins they encode and/or a lack of signal impeding the resolution of the true branching patterns, showing how previous bat phylogenies based on protein-coding genes<sup>17</sup> could have been misled<sup>34,35</sup>.